

Box-Jenkins Models

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Abstract *Time Series analysis is concerned with data that are not independent, but serially correlated, and where the relationships between consecutive observations are of interest.* O. E. Anderson, Time Series Analysis and Forecasting

1 Introduction

After we have identified and removed trends and seasonality, we assume that the residuals thus formed are a stationary time series. We then utilise particular methods to ascertain whether this residual series is the realisation of a purely random process, or if it contains serial correlation of some nature. We are going to examine the Autoregressive Integrated Moving Average Process. Another name for the processes that we will undertake is the Box-Jenkins (BJ) Methodology, which describes an iterative process for identifying a model and then using that model for forecasting. The Box-Jenkins methodology comprises four steps:

- Identification of process;
- Estimation of parameters;
- Verification, and;
- Forecasting.

Note that those four features define the original Box-Jenkins approach. We will add discussions about probabilistic forecasting and also synthetic generation of data sets.

We will be using the weather data Mildura Daily Solar Radiation and Mildura Daily Temperature for illustration, among other data sets. The first file comprises

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twelve years of daily total global solar radiation on a horizontal plane, and the second is the same period of daily average temperature. Global solar radiation means the aggregate of the solar energy hitting a site directly (direct solar radiation) plus the solar energy reflected to that site off particles in the atmosphere (diffuse solar radiation).

2 Identification of Processes

There is more than one form of a Box-Jenkins model, and we are going to investigate the Non-Seasonal Autoregressive Moving Average Model. This model is so called because it combines both the concept of a moving average model and an autoregressive model. To proceed we need some definitions and terminology. Assume we have a stationary time series X_t , ie. no trend, seasonality, periodicity, and as well we assume homocedascity (no variation in time of the variance). We will relax this subsequently and discuss the various ways the variance may change, that is either systematically or stochastically.

We thus first deal with a series that we can assume is weakly stationary.

2.1 Weak Stationarity

A series is weakly stationary if $\forall t$,

$$\begin{aligned} E[X_t] &= \mu \\ \text{Cov}[X_t, X_{t-k}] &= \gamma_k. \end{aligned}$$

2.2 Strong Stationarity

A series is said to be strongly stationary if

$$(X_1, X_2, \dots, X_m) =^d (X_{1+h}, X_{2+h}, \dots, X_{m+h}) \quad (1)$$

Here, $=^d$, means equal in distribution, or all moments are equal, not just mean and variance as with weak stationarity.

3 Autoregressive Moving Average (ARMA) Models

The general form of these models is

$$X_t - \alpha_1 X_{t-1} - \alpha_2 X_{t-2} - \dots - \alpha_p X_{t-p} = Z_t + \theta_1 Z_{t-1} + \theta_2 Z_{t-2} + \dots + \theta_q Z_{t-q} \quad (2)$$

In this the X_t are identically distributed random variables $\sim (0, \sigma_X^2)$, and the Z_t are independent and identically distributed random variables or White Noise, $\sim (0, \sigma_Z^2)$. Note that there is often the misconception that the Z_t must be Normally distributed, but for many applications, they need not be even symmetric. We shall see later when a lack of Normality is critical.

α_p and θ_q are the coefficients of p and q order polynomials satisfying

$$\begin{aligned} \alpha(y) &= 1 - \alpha_1 y - \alpha_2 y^2 - \dots - \alpha_p y^p \\ \theta(y) &= 1 + \theta_1 y + \theta_2 y^2 + \dots + \theta_q y^q, \end{aligned}$$

and $\alpha(y)$ and θ_q are the autoregressive and moving average polynomials respectively.

Let B be the backwards shift operator, defined by $B^j X_t = X_{t-j}$, $j = 0, \pm 1, \pm 2, \dots$. Then we can write Equation 2 in the form

$$\alpha(B)X_t = \theta(B)Z_t, \quad (3)$$

which is known as an $ARMA(p, q)$ process.

If $\alpha(z) = 1$, then we have a pure moving average process $MA(q)$,

$$X_t = \theta(B)Z_t. \quad (4)$$

Alternatively, if $\theta(z) = 1$, we have a pure autoregressive process $AR(p)$,

$$\alpha(B)X_t = Z_t. \quad (5)$$

4 The Sample Autocorrelation and Partial Autocorrelation Functions

The Autocorrelation and Partial Autocorrelation Functions provide a useful measure of the degree of dependence between values of a time series at specific interval of separation and thus play an important role in prediction of future values of a time series.

4.1 Digression

To fully understand these processes we need some concepts defined. We will briefly describe the concept of covariance. Suppose two variables X and Y have means μ_X , μ_Y respectively. Then the covariance of X and Y is defined to be

$$\text{Cov}(X, Y) = E(X - \mu_X)(Y - \mu_Y). \quad (6)$$

If X and Y are independent, then

$$\text{Cov}(X, Y) = E(X - \mu_X)(Y - \mu_Y) = E(X - \mu_X)E(Y - \mu_Y) = 0. \quad (7)$$

If X and Y are not independent, then the covariance may be positive or negative, depending on whether "high" values of X tend to go with "high" or "low" values of Y . It is usual to standardise the covariance by dividing by the product of their respective standard deviations to give a quantity called the correlation coefficient. If X and Y are random variables for the same stochastic process at different times, then the covariance coefficient is called an autocovariance coefficient, and the correlation coefficient is called an autocorrelation coefficient. If the process is stationary, the standard deviations of X and Y will be the same and their product will be the variance of either.

Let X_t be a stationary time series. The autocovariance function (ACVF) of X_t is given by

$$\gamma_X(h) = \text{Cov}(X_{t+h}, X_t), \quad (8)$$

and the autocorrelation function (ACF) of X_t is

$$\rho_X(h) = \frac{\gamma_X(h)}{\gamma_X(0)}. \quad (9)$$

4.2 The Sample Functions

The autocovariance and autocorrelation functions can be estimated from observations of $X_1, X_2, \dots, X_n$ to give the Sample Autocovariance Function (SAF) and the Sample Autocorrelation Function (SACF), denoted by

$$r_k = \frac{\sum_{t=1}^{n-k} (x_t - \bar{x})(x_{t+k} - \bar{x})}{\sum_{t=1}^n (x_t - \bar{x})^2} \quad (10)$$

where k is the **lag**. Thus, the SACF is a measure of the linear relationship between time series observations separated by some time period, denoted the lag k . Similar to the correlation coefficient of linear regression, r_k will take a value between $+1$ and -1 , and the closer to ± 1 the value is, the stronger the relationship. If the value is positive then the relationship is positive and vice versa.

What relationship are we exactly talking about? Let's consider a lag of 1. A value close to 1 means that there is a strong correlation between x_t and x_{t-1} , x_{t-1} and x_{t-2} and so on down to the last observation. Thus, there is a strong correlation between x_t and some x_d , where x_d is an observation quite a number of time units away from x_t . Although the correlation is strong, it does not represent the individual correlations between the variables.

The Sample Partial Autocorrelation Function describes the correlation between observations at some time period, the lag, with the influence of the serial correlation removed. The autocorrelation function gives you the connection or correlation between the data values a certain number of time steps (lags) apart. However, if the value at time t is correlated with the value at time $t - 1$ and the value at time $t - 1$ is correlated with the value at time $t - 2$, then there will be a significant correlation between the value at time t and the value at time $t - 2$ because of the interconnection. The partial autocorrelation function strips away the interconnection and gives only "pure" correlation. We can calculate the PACF coefficients knowing the ACF ones. This requires use of the Yule-Walker equations which are

$$\begin{bmatrix} 1 & \rho_1 & \rho_2 & \dots & \rho_{k-2} & \rho_{k-1} \\ \rho_1 & 1 & \rho_1 & \dots & \rho_{k-3} & \rho_{k-2} \\ \cdot & \cdot & \cdot & \dots & \cdot & \cdot \\ \cdot & \cdot & \cdot & \dots & \cdot & \cdot \\ \rho_{k-1} & \rho_{k-2} & \rho_{k-3} & \dots & \rho_1 & 1 \end{bmatrix} \begin{bmatrix} \alpha_{k1} \\ \alpha_{k2} \\ \cdot \\ \cdot \\ \alpha_{kk} \end{bmatrix} = \begin{bmatrix} \rho_1 \\ \rho_2 \\ \cdot \\ \cdot \\ \rho_k \end{bmatrix}$$

The partial autocorrelation is the α_{kk} term and they are solved progressively using

$$\alpha_{11} = \rho_1$$

$$\begin{bmatrix} 1 & \rho_1 \\ \rho_1 & 1 \end{bmatrix} \begin{bmatrix} \alpha_{21} \\ \alpha_{22} \end{bmatrix} = \begin{bmatrix} \rho_1 \\ \rho_2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & \rho_1 & \rho_2 \\ \rho_1 & 1 & \rho_1 \\ \rho_2 & \rho_1 & 1 \end{bmatrix} \begin{bmatrix} \alpha_{31} \\ \alpha_{32} \\ \alpha_{33} \end{bmatrix} = \begin{bmatrix} \rho_1 \\ \rho_2 \\ \rho_3 \end{bmatrix}$$

In fact, a recursive formula due to Durbin is more useful in estimating the coefficients.

$$\begin{aligned}
\hat{\alpha}_{p+1,j} &= \hat{\alpha}_{p,j} - \hat{\alpha}_{p+1,p+1} \hat{\alpha}_{p,p-j+1} \\
\hat{\alpha}_{p+1,p+1} &= \frac{r_{p+1} - \sum_{j=1}^p \hat{\alpha}_{p,j} r_{p+1-j}}{1 - \sum_{j=1}^p \hat{\alpha}_{p,j} r_j} \\
\hat{\alpha}_{1,1} &= r_1
\end{aligned}$$

4.3 Examples

1. For an $AR(1)$ process, $\alpha_{11} = \alpha_1 = \rho_1$, or $\hat{\alpha}_{11} = \hat{\alpha}_1 = r_1$.

2. For an $AR(2)$ process, $\begin{bmatrix} 1 & \rho_1 \\ \rho_1 & 1 \end{bmatrix} \begin{bmatrix} \alpha_{21} \\ \alpha_{22} \end{bmatrix} = \begin{bmatrix} \rho_1 \\ \rho_2 \end{bmatrix}$

$$\Rightarrow \alpha_{22} = \alpha_2 = \frac{\begin{vmatrix} 1 & \rho_1 \\ \rho_1 & \rho_2 \end{vmatrix}}{\begin{vmatrix} 1 & \rho_1 \\ \rho_1 & 1 \end{vmatrix}} = \frac{\rho_2 - \rho_1^2}{1 - \rho_1^2}.$$

3. Purely Random Process. A discrete process X_t is called a purely random process if the X_t are mutually independent, identically distributed random variables and

$$\gamma_k = \text{Cov}(X_t, X_{t+k}) = 0 \quad (11)$$

The process is strictly stationary and the ACF is given by

$$\rho_k = \begin{cases} 1, & k = 0 \\ 0, & k = \pm 1, \pm 2, \dots \end{cases} \quad (12)$$

4. Random Walk

Suppose Z_t is white noise with mean $\mu = 0$ and variance σ_Z^2 . A process X_t is denoted a random walk if

$$X_t = X_{t-1} + Z_t. \quad (13)$$

The process is customarily started at zero when $t = 0$ and so

$$\begin{aligned}
X_1 &= Z_1 \\
X_t &= \sum_{i=1}^t Z_i.
\end{aligned}$$

Thus, we find that $E(Z_t) = t\mu$ and $\text{Var}(X_t) = t\sigma_Z^2$, implying a nonstationary process. However, if we difference a random walk, we get

$$X_t - X_{t-1} = Z_t, \quad (14)$$

a purely random process.

Once we have a stationary time series we can use these functions to determine whether to fit

1. A moving average process or;
2. An autoregressive process or;
3. An autoregressive integrated moving average process

We use (1) when the SACF has spikes at lags $1, 2, \dots, q$ and the SPACF dies down gradually.

We use (2) when the SACF dies down gradually and the SPACF has spikes at lags $1, 2, \dots, p$. If neither occurs then we may be faced with a non-stationary times series or a requirement to use a combination of both, which is option (3). If we were to refer to the theoretical sample autocorrelation function (TACF) and theoretical partial autocorrelation function (TPACF), it could be shown that for the case of the $MA(1)$ process that the TACF has a non-zero correlation at lag 1 and zero autocorrelations thereafter (ie., cuts off), while the TPACF dies down in a steady manner which can be shown to be dominated by damped exponential decay. For the $AR(1)$ process, it can be shown that the TACF dies down in a damped exponential fashion while the TPACF has a non zero partial autocorrelation at lag 1, with zeros thereafter, that is, the TPAC cuts off. The graphs produced from the SACF are known as correlograms.

5 Moving Average Process

A moving average process $MA(q)$ (of order q) has the present value of the series written as a weighted sum of or regression on past random shocks,

$$X_t = Z_t + \theta_1 Z_{t-1} + \theta_2 Z_{t-2} + \dots + \theta_q Z_{t-q}, \quad (15)$$

where $Z_t \sim (0, \sigma_Z^2)$.

We find that

$$E(X_t) = 0$$

$$\sigma_X^2 = \sigma_Z^2 \left(1 + \sum_{i=1}^q \theta_i^2 \right)$$

6 Autoregressive Processes

The general form of an autoregressive process of order p , or $AR(p)$, is given by

$$X_t = \alpha_1 X_{t-1} + \alpha_2 X_{t-2} + \dots + \alpha_p X_{t-p} + Z_t. \quad (16)$$

This is like a multiple regression but X_t is regressed on past values of X_t , not on other predictor variables, hence the term autoregressive. Remember that another way to write this process is (using the backward shift operator B):

$$\alpha(B)X_t = Z_t, \tag{17}$$

where $\alpha(B) = 1 - \alpha_1 B - \alpha_2 B^2 - \dots - \alpha_p B^p$ is called the $AR(p)$ operator.